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324513 (24)

BE (5th Semester)

Examination, April-May, 2014

Branch : Electrical

ANALOG ELECTRONICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory. Attempt any two parts from (b), (c) & (d) of each question.

Q. 1. (a) Define f_T. 2

(b) Derive the expression for current gain with resistive load. 7

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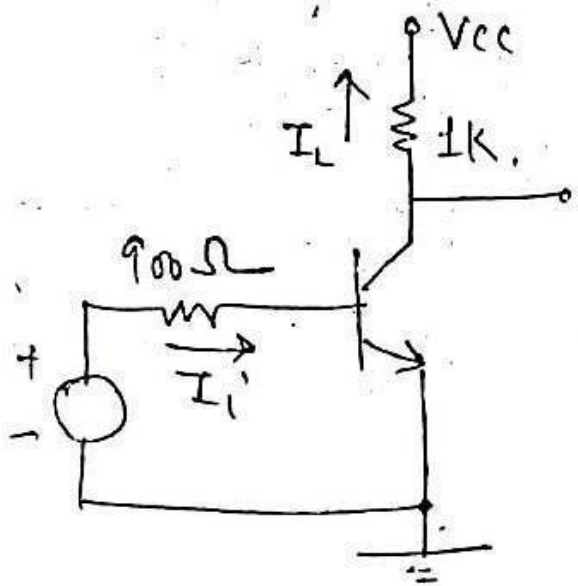
(c) Using Miller's theorem & approximate analysis, compute : 7

(i) The upper 3 dB frequency of current gain

(ii) The magnitude of the voltage gain $A_{vs} = V_0/V_s$ at frequency of the part (i)



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(d) Given the following transistor measurements made at I_C = 5 mA, V_{ce} = 10 V and at room temperature h_{fe} = 100, h_{ie} = 600 Ω, C_c = 3 pF, A_i = 10 at f = 10 MHz. Find f_β, f_T, $C_{e'}$, $r_{b'e}$ & $r_{bb'}$. 7

Q. 2. (a) What is the remedy of cross-over distortion ? 2

(b) Find the conversion efficiency of transformer coupled class B-push-pull amplifier. 7

(c) For a transformer-coupled class-A amplifier prove that conversion efficiency η = 50%. 7

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(d) For a transformer coupled push-pull clas amplifier prove that even hormonics eliminated.

Q. 3. (a) Find the rise-time, if f_H = 1 MHz & f_L = 1k Also find bandwidth of amplifier.

(b) Derive the expression for low-frequs response of an amplifier. Also define f_L.

(c) Derive the expression for the high 3 frequency f_H^* of n identical non-interac stages in terms of f_H for one stage.

(d) Define the following :

(i) Rise time

(ii) Tilt or sag

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Q. 4. (a) For a signal $v(t) = 5 \cos \omega t$ find slew rate if maximum frequency $f_{max} = 10$ kHz 2

(b) What are the applications of OPAMP. Explain any one. 7

(c) Draw the block diagram of OPAMP and explain each block with suitable circuit diagram. 7

(d) Explain bridge amplifier using OPAMP. 7

Q. 5. (a) What are the advantages of integrated circuits ? 2

(b) List the steps involved in fabricating a monolitic integrated circuit and explain. 7

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(c) Explain monolithic diodes. 7

(d) Explain wet and dry itching. 7

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